COGNIZANT DIGITAL NURTURE 5.0

JAVA FSE DEEP SKILLING ASSESSMENTS

Week 5 Assessment Report

Topic: React JS (Hands-on Labs 1 - 17)

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GitHub Repository: https://github.com/ishanparmar2105-stack/digital-nurture-project

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1. Topic Overview & Architecture

This solution contains the implementation of the Week 5 React JS Hands-on Labs (HOL 1 to 17) for the Cognizant Digital Nurture program.

To deliver a clean, enterprise-grade submission, we developed a unified Single Page Application (SPA) dashboard rather than 17 disconnected directories. The dashboard has a responsive navigation sidebar and a central workspace shell. It compiles all labs in a single React tree, ensuring fast and isolated component renders.

Core React features demonstrated in this project include:
- Rendering JSX elements and configuring attributes
- Functional and Class-based components
- Prop drilling vs React Context API (Theme Context)
- Event handling, synthetics, and argument binding
- Conditional rendering (ternary, logic AND, element variables)
- State management (increment/decrement loops, arrays, objects)
- Form controls (controlled inputs, submission alerts, validations)
- Client Routing (using react-router-dom for nested routing)
- Async API Integration (fetching random user data in componentDidMount)

- Styling methodologies (Vanilla CSS, CSS Modules, inline dynamic logic)

2. Core Configuration & Layout Entrypoints

2.1. package.json

// File: package.json

```
{
  "name": "react-deepskilling-apps",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "oxlint",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.7",
    "react-dom": "^19.2.7",
    "react-router-dom": "^7.18.1"
  },
  "devDependencies": {
    "@types/react": "^19.2.17",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.3",
    "oxlint": "^1.71.0",
    "vite": "^8.1.1"
  }
}
```

2.2. vite.config.js

// File: vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

2.3. src/main.jsx

// File: src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

2.4. src/App.jsx (Unified Dashboard Sidebar shell)

// File: src/App.jsx

```
import React, { useState } from 'react';
import HOL1Welcome from './Components/HOL1Welcome';
import HOL2StudentPortal from './Components/HOL2StudentPortal';
import HOL3ScoreCalculator from './Components/HOL3ScoreCalculator';
import HOL4PostsFetch from './Components/HOL4PostsFetch';
import HOL5CohortDetails from './Components/HOL5CohortDetails';
import HOL6TrainersRouter from './Components/HOL6TrainersRouter';
import HOL7PropsShopping from './Components/HOL7PropsShopping';
import HOL8CounterApp from './Components/HOL8CounterApp';
import HOL9CricketPlayers from './Components/HOL9CricketPlayers';
import HOL10OfficeSpace from './Components/HOL10OfficeSpace';
import HOL11EventExamples from './Components/HOL11EventExamples';
import HOL12LoginLogout from './Components/HOL12LoginLogout';
import HOL13BloggerApp from './Components/HOL13BloggerApp';
import HOL14ContextAPI from './Components/HOL14ContextAPI';
import HOL15TicketRegister from './Components/HOL15TicketRegister';
import HOL16MailRegister from './Components/HOL16MailRegister';
import HOL17FetchUser from './Components/HOL17FetchUser';

const labs = [
  { id: 'hol1', title: 'HOL 1: Welcome Page', component: <HOL1Welcome /> },
  { id: 'hol2', title: 'HOL 2: Student Portal', component: <HOL2StudentPortal /> },
  { id: 'hol3', title: 'HOL 3: Score Calculator', component: <HOL3ScoreCalculator /> },
  { id: 'hol4', title: 'HOL 4: Fetch Posts API', component: <HOL4PostsFetch /> },
  { id: 'hol5', title: 'HOL 5: Cohort Details', component: <HOL5CohortDetails /> },
  { id: 'hol6', title: 'HOL 6: Trainers SPA', component: <HOL6TrainersRouter /> },
  { id: 'hol7', title: 'HOL 7: Props Shopping', component: <HOL7PropsShopping /> },
  { id: 'hol8', title: 'HOL 8: State Counter', component: <HOL8CounterApp /> },
  { id: 'hol9', title: 'HOL 9: ES6 Cricket App', component: <HOL9CricketPlayers /> },
  { id: 'hol10', title: 'HOL 10: Office Space JSX', component: <HOL10OfficeSpace /> },
  { id: 'hol11', title: 'HOL 11: Event Examples', component: <HOL11EventExamples /> },
  { id: 'hol12', title: 'HOL 12: Conditional Login', component: <HOL12LoginLogout /> },
  { id: 'hol13', title: 'HOL 13: Blogger App', component: <HOL13BloggerApp /> },
  { id: 'hol14', title: 'HOL 14: Context API Theme', component: <HOL14ContextAPI /> },
  { id: 'hol15', title: 'HOL 15: Ticket Register', component: <HOL15TicketRegister /> },
  { id: 'hol16', title: 'HOL 16: Mail Registration', component: <HOL16MailRegister /> },
  { id: 'hol17', title: 'HOL 17: Fetch Async User', component: <HOL17FetchUser /> }
];

export default function App() {
  const [activeLabId, setActiveLabId] = useState('hol1');
```

```
  const activeLab = labs.find(lab => lab.id === activeLabId) || labs[0];

  return (
    <div style={{ display: 'flex', height: '100vh', width: '100vw', fontFamily: '"Outfit", "Inter",
sans-serif', background: '#f7fafc', overflow: 'hidden' }}>

      {/* Sidebar Navigation */}
      <aside style={{ width: '300px', background: '#2d3748', display: 'flex', flexDirection: 'column', color:
'#fff', borderRight: '1px solid #4a5568' }}>
        <div style={{ padding: '20px', borderBottom: '1px solid #4a5568', background: '#1a202c' }}>
          <h2 style={{ fontSize: '18px', margin: 0, fontWeight: '800', tracking: 'wide', color: '#63b3ed' }}>
            React JS Deep Skilling
          </h2>
          <span style={{ fontSize: '11px', color: '#a0aec0', display: 'block', marginTop: '4px' }}>
            Candidate: Ishan Parmar (ID: 12239022)
          </span>
        </div>

        {/* Labs Scrollable List */}
        <nav style={{ flex: 1, overflowY: 'auto', padding: '15px 10px' }}>
          <div style={{ display: 'flex', flexDirection: 'column', gap: '5px' }}>
            {labs.map(lab => {
              const isActive = lab.id === activeLabId;
              return (
                <button
                  key={lab.id}
                  id={`btn-${lab.id}`}
                  onClick={() => setActiveLabId(lab.id)}
                  style={{
                    display: 'block',
                    width: '100%',
                    textAlign: 'left',
                    padding: '10px 14px',
                    border: 'none',
                    borderRadius: '6px',
                    cursor: 'pointer',
                    fontSize: '13px',
                    fontWeight: isActive ? 'bold' : '500',
                    background: isActive ? '#3182ce' : 'transparent',
                    color: isActive ? '#fff' : '#cbd5e0',
                    transition: 'all 0.15s ease'
                  }}
                >
                  {lab.title}
                </button>
              );
            })}
          </div>
        </nav>

        <div style={{ padding: '15px', background: '#1a202c', borderTop: '1px solid #4a5568', textAlign:
'center', fontSize: '11px', color: '#718096' }}>
          Week 5 Evaluation Workspace
        </div>
      </aside>

      {/* Main Workspace display */}
      <main style={{ flex: 1, display: 'flex', flexDirection: 'column', overflow: 'hidden' }}>
        {/* Header */}
        <header style={{ height: '60px', background: '#fff', borderBottom: '1px solid #e2e8f0', display:
'flex', alignItems: 'center', padding: '0 30px', justifyContent: 'space-between', flexShrink: 0 }}>
          <h3 style={{ color: '#2d3748', fontSize: '16px', margin: 0, fontWeight: '700' }}>
```

```
            {activeLab.title}
          </h3>
          <span style={{ fontSize: '12px', background: '#ebf8ff', color: '#2b6cb0', padding: '4px 10px',
borderRadius: '20px', fontWeight: 'bold' }}>
            Active Session
          </span>
        </header>

        {/* Content body */}
        <div style={{ flex: 1, overflowY: 'auto', padding: '30px', background: '#f7fafc' }}>
          <div style={{ maxWidth: '900px', margin: '0 auto' }}>
            {activeLab.component}
          </div>
        </div>
      </main>

    </div>
  );
}
```

2.5. Global Stylesheets & ThemeContext

// File: src/index.css

```
/* Modern CSS Reset & Core Design System */
@import url('https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;500;600;700;800&display=swap');

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  margin: 0;
  font-family: 'Outfit', 'Inter', system-ui, -apple-system, sans-serif;
  background-color: #f7fafc;
  color: #2d3748;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;
}

#root {
  width: 100vw;
  height: 100vh;
  margin: 0;
  padding: 0;
  text-align: left;
  border: none;
  min-height: auto;
  display: block;
}

/* Custom Scrollbars */
::-webkit-scrollbar {
  width: 6px;
  height: 6px;
}

::-webkit-scrollbar-track {
  background: #f1f1f1;
}

::-webkit-scrollbar-thumb {
  background: #cbd5e0;
  border-radius: 4px;
}

::-webkit-scrollbar-thumb:hover {
  background: #a0aec0;
}
```

// File: src/Components/ThemeContext.js

```
import React from 'react';

// Create ThemeContext with default value 'light'
const ThemeContext = React.createContext('light');

export default ThemeContext;
```

// File: src/Stylesheets/mystyle.css

```
/* mystyle.css for score calculator and other components */
.score-calculator-card {
  border: 1px solid #e2e8f0;
  border-radius: 8px;
  padding: 20px;
  background-color: #f7fafc;
  max-width: 450px;
  margin: 15px auto;
  box-shadow: 0 4px 6px rgba(0, 0, 0, 0.05);
}

.score-calculator-title {
  color: #2b6cb0;
  text-align: center;
  margin-bottom: 15px;
  font-size: 20px;
  font-weight: bold;
}

.score-details-list {
  list-style: none;
  padding: 0;
  margin: 0 0 15px 0;
}

.score-details-item {
  display: flex;
  justify-content: space-between;
  padding: 8px 0;
  border-bottom: 1px dashed #e2e8f0;
  font-size: 14px;
}

.score-details-label {
  font-weight: 600;
  color: #4a5568;
}

.score-details-value {
  color: #2d3748;
}

.score-result-badge {
  text-align: center;
  padding: 10px;
  border-radius: 6px;
  font-weight: bold;
  font-size: 15px;
  margin-top: 10px;
}

.score-result-badge.met {
  background-color: #c6f6d5;
  color: #22543d;
  border: 1px solid #9ae6b4;
}

.score-result-badge.unmet {
  background-color: #fed7d7;
  color: #742a2a;
```

```
  border: 1px solid #feb2b2;
}
```

HOL 1: Welcome Message

Objective: Render a simple welcome heading on the page.

// File: HOL1Welcome.jsx

```
import React from 'react';

export default function HOL1Welcome() {
  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 1: Welcome Page</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Set up react environment and render a welcome heading.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />
      <h1 id="welcome-heading" style={{ color: '#1a202c', textAlign: 'center', margin: '30px 0', fontSize: '28px', fontWeight: 'bold' }}>
        welcome to the first session of React
      </h1>
    </div>
  );
}
```

HOL 2: Student Portal

Objective: Create Home, About, and Contact sub-components inside a main portal component.

// File: HOL2StudentPortal.jsx

```
import React from 'react';

// Sub-components
class Home extends React.Component {
  render() {
    return (
      <div style={{ padding: '15px', background: '#ebf8ff', borderLeft: '4px solid #3182ce', margin: '10px 0',
borderRadius: '4px' }}>
        <h3 style={{ margin: '0 0 5px 0', color: '#2b6cb0' }}>Home Component</h3>
        <p style={{ margin: 0, color: '#2d3748' }}>Welcome to the Home page of Student Management Portal</p>
      </div>
    );
  }
}

class About extends React.Component {
  render() {
    return (
      <div style={{ padding: '15px', background: '#f0fff4', borderLeft: '4px solid #38a169', margin: '10px 0',
borderRadius: '4px' }}>
        <h3 style={{ margin: '0 0 5px 0', color: '#276749' }}>About Component</h3>
        <p style={{ margin: 0, color: '#2d3748' }}>Welcome to the About page of the Student Management
Portal</p>
      </div>
    );
  }
}

class Contact extends React.Component {
  render() {
    return (
      <div style={{ padding: '15px', background: '#fffaf0', borderLeft: '4px solid #dd6b20', margin: '10px 0',
borderRadius: '4px' }}>
        <h3 style={{ margin: '0 0 5px 0', color: '#9c4221' }}>Contact Component</h3>
        <p style={{ margin: 0, color: '#2d3748' }}>Welcome to the Contact page of the Student Management
Portal</p>
      </div>
    );
  }
}

export default function HOL2StudentPortal() {
  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 2: Student
Management Portal</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Create multiple class components and render them within App shell.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />
      <div style={{ display: 'flex', flexDirection: 'column', gap: '10px' }}>
        <Home />
        <About />
        <Contact />
```

```
      </div>
    </div>
  );
}
```

HOL 3: Score Calculator

Objective: Pass Name, School, Total, Goal props to a functional component and calculate average score with external stylesheet.

// File: HOL3ScoreCalculator.jsx

```
import React, { useState } from 'react';
import '../Stylesheets/mystyle.css';

// The CalculateScore component as specified in the HOL
export function CalculateScore({ Name, School, Total, goal }) {
  // Let's assume the average is Total / 4 (for 4 subjects)
  const average = (Total / 4).toFixed(2);
  const isGoalMet = parseFloat(average) >= parseFloat(goal);

  return (
    <div className="score-calculator-card">
      <div className="score-calculator-title">Student Score Report</div>
      <ul className="score-details-list">
        <li className="score-details-item">
          <span className="score-details-label">Student Name:</span>
          <span className="score-details-value">{Name}</span>
        </li>
        <li className="score-details-item">
          <span className="score-details-label">School:</span>
          <span className="score-details-value">{School}</span>
        </li>
        <li className="score-details-item">
          <span className="score-details-label">Total Score (out of 400):</span>
          <span className="score-details-value">{Total}</span>
        </li>
        <li className="score-details-item">
          <span className="score-details-label">Target Goal (Average):</span>
          <span className="score-details-value">{goal}</span>
        </li>
        <li className="score-details-item" style={{ borderBottom: 'none', fontSize: '16px', fontWeight: 'bold'
}}>
          <span className="score-details-label" style={{ color: '#2b6cb0' }}>Average Calculated:</span>
          <span className="score-details-value" style={{ color: '#2b6cb0' }}>{average}</span>
        </li>
      </ul>
      <div className={`score-result-badge ${isGoalMet ? 'met' : 'unmet'}`}>
        {isGoalMet ? '[MET] TARGET GOAL MET' : '[UNMET] TARGET GOAL NOT MET'}
      </div>
    </div>
  );
}

export default function HOL3ScoreCalculator() {
  const [name, setName] = useState('Ishan Parmar');
  const [school, setSchool] = useState('Cognizant Academy');
  const [total, setTotal] = useState(360);
  const [goal, setGoal] = useState(85);

  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 3: Score
Calculator</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
```

```
        Objective: Create a functional component accepting props to perform calculations, styled using external
CSS.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />

      {/* Interactive controls */}
      <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(180px, 1fr))', gap: '15px',
marginBottom: '20px', padding: '15px', background: '#f7fafc', borderRadius: '6px' }}>
        <div>
          <label style={{ display: 'block', fontSize: '12px', fontWeight: 'bold', color: '#4a5568',
marginBottom: '4px' }}>Name</label>
          <input type="text" value={name} onChange={(e) => setName(e.target.value)} style={{ width: '100%',
padding: '6px', border: '1px solid #cbd5e0', borderRadius: '4px' }} />
        </div>
        <div>
          <label style={{ display: 'block', fontSize: '12px', fontWeight: 'bold', color: '#4a5568',
marginBottom: '4px' }}>School</label>
          <input type="text" value={school} onChange={(e) => setSchool(e.target.value)} style={{ width: '100%',
padding: '6px', border: '1px solid #cbd5e0', borderRadius: '4px' }} />
        </div>
        <div>
          <label style={{ display: 'block', fontSize: '12px', fontWeight: 'bold', color: '#4a5568',
marginBottom: '4px' }}>Total (out of 400)</label>
          <input type="number" value={total} onChange={(e) => setTotal(Number(e.target.value))} style={{ width:
'100%', padding: '6px', border: '1px solid #cbd5e0', borderRadius: '4px' }} />
        </div>
        <div>
          <label style={{ display: 'block', fontSize: '12px', fontWeight: 'bold', color: '#4a5568',
marginBottom: '4px' }}>Target Goal (0-100)</label>
          <input type="number" value={goal} onChange={(e) => setGoal(Number(e.target.value))} style={{ width:
'100%', padding: '6px', border: '1px solid #cbd5e0', borderRadius: '4px' }} />
        </div>
      </div>

      <CalculateScore Name={name} School={school} Total={total} goal={goal} />
    </div>
  );
}
```

HOL 4: Class Component & Fetch

Objective: Fetch posts from JSONPlaceholder on mount and display them using a Class component.

// File: HOL4PostsFetch.jsx

```
import React from 'react';

export class Posts extends React.Component {
  constructor(props) {
    super(props);
    this.state = {
      posts: [],
      loading: true,
      error: null
    };
  }

  loadPosts() {
    fetch('https://jsonplaceholder.typicode.com/posts')
      .then(response => {
        if (!response.ok) {
          throw new Error('Failed to fetch posts');
        }
        return response.json();
      })
      .then(data => {
        // limit to first 6 posts for cleaner UI display
        this.setState({ posts: data.slice(0, 6), loading: false });
      })
      .catch(error => {
        this.setState({ error: error.message, loading: false });
      });
  }

  componentDidMount() {
    this.loadPosts();
  }

  render() {
    const { posts, loading, error } = this.state;

    if (loading) {
      return (
        <div style={{ textAlign: 'center', padding: '20px', color: '#4a5568', fontWeight: 'bold' }}>
          Loading posts from REST API...
        </div>
      );
    }

    if (error) {
      return (
        <div style={{ color: '#e53e3e', padding: '15px', background: '#fff5f5', borderRadius: '6px', border:
'1px solid #fed7d7' }}>
          Error loading posts: {error}
        </div>
      );
    }

    return (
      <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(280px, 1fr))', gap: '15px'
```

```
}}>
        {posts.map(post => (
          <div key={post.id} style={{ border: '1px solid #e2e8f0', borderRadius: '6px', padding: '15px',
background: '#fff', boxShadow: '0 2px 4px rgba(0,0,0,0.02)' }}>
            <span style={{ fontSize: '11px', fontWeight: 'bold', background: '#e2e8f0', color: '#4a5568',
padding: '2px 6px', borderRadius: '4px' }}>
              Post ID: {post.id}
            </span>
            <h4 style={{ margin: '8px 0', color: '#2c5282', fontSize: '15px', textTransform: 'capitalize' }}>
              {post.title}
            </h4>
            <p style={{ margin: 0, color: '#4a5568', fontSize: '13px', lineHeight: '1.4' }}>
              {post.body}
            </p>
          </div>
        ))}
      </div>
    );
  }
}

export default function HOL4PostsFetch() {
  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 4: Lifecycle Hooks
& Fetch API</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Create a class component that fetches data using Fetch API in componentDidMount.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />
      <Posts />
    </div>
  );
}
```

HOL 5: CSS Module Cohort

Objective: Use CSS modules to style elements and dynamically color text based on cohort status.

// File: HOL5CohortDetails.jsx

```
import React, { useState } from 'react';
import styles from './CohortDetails.module.css';

export function CohortDetails({ cohortCode, cohortName, status }) {
  // Define dynamic style for h3 element
  const statusColor = status === 'ongoing' ? 'green' : 'blue';

  return (
    <div className={styles.box}>
      <h3 style={{ color: statusColor, margin: '0 0 10px 0', fontSize: '18px', textTransform: 'capitalize' }}>
        Status: {status}
      </h3>
      <dl>
        <dt>Cohort Code</dt>
        <dd>{cohortCode}</dd>
        <dt>Cohort Name</dt>
        <dd>{cohortName}</dd>
      </dl>
    </div>
  );
}

export default function HOL5CohortDetails() {
  const [cohorts, setCohorts] = useState([
    { id: 1, code: 'ADM23JA001', name: 'Java Full Stack Engineering', status: 'ongoing' },
    { id: 2, code: 'ADM23CS004', name: 'Cloud & DevOps Solutions', status: 'completed' },
    { id: 3, code: 'ADM23DA009', name: 'Data Analytics & AI Engineering', status: 'upcoming' }
  ]);

  const toggleStatus = (id) => {
    setCohorts(cohorts.map(c => {
      if (c.id === id) {
        const nextStatus = c.status === 'ongoing' ? 'completed' : c.status === 'completed' ? 'upcoming' :
'ongoing';
        return { ...c, status: nextStatus };
      }
      return c;
    }));
  };

  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 5: Styling
Components (CSS Modules)</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Apply CSS Modules styling, style tag selectors, and handle dynamic styling conditionally
based on status.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />

      <p style={{ fontSize: '13px', color: '#4a5568', marginBottom: '15px' }}>
        ? <em>Click any cohort card to cycle its status and observe the dynamic header color
(<strong>green</strong> when ongoing, <strong>blue</strong> otherwise).</em>
      </p>
```

```
      <div style={{ display: 'flex', flexWrap: 'wrap', gap: '15px', justifyContent: 'center' }}>
        {cohorts.map(c => (
          <div key={c.id} onClick={() => toggleStatus(c.id)} style={{ cursor: 'pointer' }}>
            <CohortDetails cohortCode={c.code} cohortName={c.name} status={c.status} />
          </div>
        ))}
      </div>
    </div>
  );
}
```

// File: CohortDetails.module.css

```
/* CohortDetails.module.css */
.box {
  width: 300px;
  display: inline-block;
  margin: 10px;
  padding-top: 10px;
  padding-bottom: 10px;
  padding-left: 20px;
  padding-right: 20px;
  border: 1px solid #000000;
  border-radius: 10px;
  background-color: #ffffff;
  box-shadow: 0 4px 6px rgba(0, 0, 0, 0.02);
  vertical-align: top;
}

.box dt {
  font-weight: 500;
  color: #4a5568;
  margin-top: 6px;
}

.box dd {
  margin-left: 0;
  margin-bottom: 6px;
  color: #1a202c;
  font-size: 14px;
}
```

HOL 6: Trainers SPA React Router

Objective: Integrate client routing to display list of trainers and show details using Route parameters.

// File: HOL6TrainersRouter.jsx

```
import React from 'react';
import { MemoryRouter, Routes, Route, Link, useParams, useNavigate } from 'react-router-dom';

// 1. Mock Data (TrainersMock.js equivalent)
const trainersMockData = [
  { TrainerId: 'T101', Name: 'Suresh Kumar', Email: 'suresh@cognizant.com', Phone: '9876543210', Technology:
'Java FSE', Skills: 'Spring Boot, Microservices, Hibernate' },
  { TrainerId: 'T102', Name: 'Anjali Sharma', Email: 'anjali@cognizant.com', Phone: '9876543211', Technology:
'React JS', Skills: 'React, Redux, Webpack, ES6' },
  { TrainerId: 'T103', Name: 'Robert Miller', Email: 'robert@cognizant.com', Phone: '9876543212', Technology:
'Cloud Engineering', Skills: 'AWS, Kubernetes, Terraform, Docker' },
  { TrainerId: 'T104', Name: 'Priya Patel', Email: 'priya@cognizant.com', Phone: '9876543213', Technology:
'Data Analytics', Skills: 'Python, Spark, Hadoop, SQL' }
];

// 2. Home Component
function Home() {
  return (
    <div style={{ padding: '15px', background: '#f7fafc', borderRadius: '6px', border: '1px solid #e2e8f0' }}>
      <h3 style={{ color: '#2b6cb0', marginTop: 0 }}>Cognizant Academy Trainer Portal</h3>
      <p style={{ fontSize: '14px', color: '#4a5568', lineHeight: '1.5' }}>
        Welcome to the Academy Trainer Directory. This Single Page Application helps management teams maintain
lists of specialized trainers and view their profiles, streams, and core technologies.
      </p>
      <div style={{ marginTop: '15px' }}>
        <Link to="/trainers" style={{ display: 'inline-block', background: '#3182ce', color: '#fff', padding:
'8px 16px', borderRadius: '4px', textDecoration: 'none', fontWeight: 'bold', fontSize: '13px' }}>
          Explore Trainers List ?
        </Link>
      </div>
    </div>
  );
}

// 3. TrainersList Component
function TrainersList() {
  return (
    <div>
      <h3 style={{ color: '#2b6cb0', marginTop: 0 }}>Trainer Directory</h3>
      <p style={{ fontSize: '13px', color: '#718096', marginBottom: '15px' }}>
        Select a trainer name to view their full competence profile:
      </p>
      <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
        {trainersMockData.map(t => (
          <Link key={t.TrainerId} to={`/trainers/${t.TrainerId}`} style={{ display: 'block', padding: '12px',
background: '#fff', border: '1px solid #e2e8f0', borderRadius: '6px', textDecoration: 'none', color: '#2d3748',
transition: 'background 0.2s' }}>
            <div style={{ display: 'flex', justifyContent: 'between', alignItems: 'center' }}>
              <div>
                <strong style={{ color: '#2c5282' }}>{t.Name}</strong>
                <span style={{ fontSize: '12px', color: '#718096', marginLeft: '10px'
}}>({t.Technology})</span>
              </div>
              <div style={{ fontSize: '12px', color: '#3182ce', fontWeight: '600', marginLeft: 'auto' }}>
                View Profile &gt;
```

```
              </div>
            </div>
          </Link>
        ))}
      </div>
    </div>
  );
}

// 4. TrainerDetail Component
function TrainerDetail() {
  const { id } = useParams();
  const navigate = useNavigate();
  const trainer = trainersMockData.find(t => t.TrainerId === id);

  if (!trainer) {
    return (
      <div style={{ padding: '15px', color: '#e53e3e', background: '#fff5f5', borderRadius: '6px' }}>
        Trainer profile with ID "{id}" was not found in the directory database.
      </div>
    );
  }

  return (
    <div style={{ padding: '20px', background: '#fff', border: '1px solid #e2e8f0', borderRadius: '8px' }}>
      <div style={{ display: 'flex', justifyContent: 'between', alignItems: 'center', marginBottom: '15px' }}>
        <h3 style={{ color: '#2b6cb0', margin: 0 }}>Trainer Profile: {trainer.Name}</h3>
        <button onClick={() => navigate('/trainers')} style={{ marginLeft: 'auto', background: '#edf2f7',
border: '1px solid #cbd5e0', padding: '4px 10px', borderRadius: '4px', cursor: 'pointer', fontSize: '12px',
fontWeight: '600' }}>
          Back to List
        </button>
      </div>
      <table style={{ width: '100%', borderCollapse: 'collapse', fontSize: '13px' }}>
        <tbody>
          <tr style={{ borderBottom: '1px solid #edf2f7' }}>
            <td style={{ padding: '8px 0', fontWeight: 'bold', color: '#4a5568', width: '120px' }}>Trainer
ID:</td>
            <td style={{ padding: '8px 0', color: '#2d3748' }}>{trainer.TrainerId}</td>
          </tr>
          <tr style={{ borderBottom: '1px solid #edf2f7' }}>
            <td style={{ padding: '8px 0', fontWeight: 'bold', color: '#4a5568' }}>Email:</td>
            <td style={{ padding: '8px 0', color: '#2d3748' }}>{trainer.Email}</td>
          </tr>
          <tr style={{ borderBottom: '1px solid #edf2f7' }}>
            <td style={{ padding: '8px 0', fontWeight: 'bold', color: '#4a5568' }}>Phone:</td>
            <td style={{ padding: '8px 0', color: '#2d3748' }}>{trainer.Phone}</td>
          </tr>
          <tr style={{ borderBottom: '1px solid #edf2f7' }}>
            <td style={{ padding: '8px 0', fontWeight: 'bold', color: '#4a5568' }}>Technology:</td>
            <td style={{ padding: '8px 0', color: '#2d3748' }}><span style={{ background: '#ebf8ff', color:
'#2b6cb0', padding: '2px 6px', borderRadius: '4px', fontWeight: '600' }}>{trainer.Technology}</span></td>
          </tr>
          <tr>
            <td style={{ padding: '8px 0', fontWeight: 'bold', color: '#4a5568' }}>Core Skills:</td>
            <td style={{ padding: '8px 0', color: '#2d3748' }}>{trainer.Skills}</td>
          </tr>
        </tbody>
      </table>
    </div>
  );
}
```

```
// 5. Main Component Layout
export default function HOL6TrainersRouter() {
  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px 
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 6: React Router 
Single Page Application</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Set up routing, dynamic path resolution, route navigation link mapping, and useParams hooks.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />

      <MemoryRouter initialEntries={['/']}>
        {/* Navigation bar of nested router */}
        <nav style={{ display: 'flex', gap: '15px', background: '#ebf8ff', padding: '10px 15px', borderRadius: 
'6px', marginBottom: '20px' }}>
          <Link to="/" style={{ color: '#2c5282', textDecoration: 'none', fontWeight: 'bold', fontSize: '13px' 
}}>Home</Link>
          <span style={{ color: '#a0aec0' }}>|</span>
          <Link to="/trainers" style={{ color: '#2c5282', textDecoration: 'none', fontWeight: 'bold', fontSize: 
'13px' }}>Trainers List</Link>
        </nav>

        {/* Routes configuration */}
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/trainers" element={<TrainersList />} />
          <Route path="/trainers/:id" element={<TrainerDetail />} />
        </Routes>
      </MemoryRouter>
    </div>
  );
}
```

HOL 7: Props Shopping

Objective: Props passing from parent component to child component to display cart items.

// File: HOL7PropsShopping.jsx

```
import React from 'react';

// The Cart component representing a single item row/card receiving props
class Cart extends React.Component {
  render() {
    const { Itemname, Price } = this.props;
    return (
      <tr style={{ borderBottom: '1px solid #edf2f7' }}>
        <td style={{ padding: '10px', textAlign: 'left', color: '#2d3748', fontWeight: '500' }}>{Itemname}</td>
        <td style={{ padding: '10px', textAlign: 'right', color: '#4a5568', fontFamily: 'monospace'
}}>${Price.toFixed(2)}</td>
      </tr>
    );
  }
}

// The OnlineShopping parent component holding list of items and passing props
export class OnlineShopping extends React.Component {
  render() {
    // Array of 5 items
    const items = [
      { id: 1, name: 'MacBook Air M3', price: 1099.00 },
      { id: 2, name: 'iPhone 15 Pro', price: 999.00 },
      { id: 3, name: 'AirPods Pro 2', price: 249.00 },
      { id: 4, name: 'iPad Pro 11-inch', price: 799.00 },
      { id: 5, name: 'Apple Watch Series 9', price: 399.00 }
    ];

    const totalPrice = items.reduce((sum, item) => sum + item.price, 0);

    return (
      <div style={{ maxWidth: '500px', margin: '0 auto', background: '#fff', border: '1px solid #e2e8f0',
borderRadius: '8px', padding: '20px', boxShadow: '0 4px 6px rgba(0,0,0,0.02)' }}>
        <h3 style={{ color: '#2b6cb0', textAlign: 'center', margin: '0 0 15px 0' }}>Online Shopping
Invoice</h3>
        <table style={{ width: '100%', borderCollapse: 'collapse' }}>
          <thead>
            <tr style={{ background: '#f7fafc', borderBottom: '2px solid #cbd5e0' }}>
              <th style={{ padding: '10px', textAlign: 'left', fontSize: '12px', color: '#4a5568',
textTransform: 'uppercase' }}>Item Name</th>
              <th style={{ padding: '10px', textAlign: 'right', fontSize: '12px', color: '#4a5568',
textTransform: 'uppercase' }}>Price</th>
            </tr>
          </thead>
          <tbody>
            {items.map(item => (
              <Cart key={item.id} Itemname={item.name} Price={item.price} />
            ))}
            <tr style={{ fontWeight: 'bold', background: '#f7fafc', fontSize: '15px' }}>
              <td style={{ padding: '12px 10px', color: '#2d3748' }}>Grand Total:</td>
              <td style={{ padding: '12px 10px', textAlign: 'right', color: '#2b6cb0', fontFamily: 'monospace'
}}>${totalPrice.toFixed(2)}</td>
            </tr>
          </tbody>
        </table>
```

```
      </div>
    );
  }
}

export default function HOL7PropsShopping() {
  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 7: Working with
Component Props</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Implement parent-child prop rendering using class components.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />
      <OnlineShopping />
    </div>
  );
}
```

HOL 8: CounterApp CountPeople

Objective: Track person entry/exit using Class component local state.

// File: HOL8CounterApp.jsx

```
import React from 'react';

export class CountPeople extends React.Component {
  constructor(props) {
    super(props);
    // Initialize entrycount and exitcount
    this.state = {
      entrycount: 0,
      exitcount: 0
    };

    // Bind event handlers
    this.updateEntry = this.updateEntry.bind(this);
    this.updateExit = this.updateExit.bind(this);
  }

  updateEntry() {
    this.setState(prevState => ({
      entrycount: prevState.entrycount + 1
    }));
  }

  updateExit() {
    this.setState(prevState => ({
      exitcount: prevState.exitcount + 1
    }));
  }

  render() {
    const { entrycount, exitcount } = this.state;
    const currentPeople = entrycount - exitcount;

    return (
      <div style={{ maxWidth: '400px', margin: '0 auto', border: '1px solid #e2e8f0', borderRadius: '8px',
padding: '20px', background: '#fff', boxShadow: '0 4px 6px rgba(0,0,0,0.02)' }}>
        <h3 style={{ color: '#2b6cb0', textAlign: 'center', margin: '0 0 20px 0' }}>Mall Footfall Tracker</h3>

        <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '15px', marginBottom: '20px' }}>
          <div style={{ background: '#ebf8ff', padding: '15px', borderRadius: '6px', textAlign: 'center' }}>
            <span style={{ fontSize: '12px', fontWeight: 'bold', color: '#2b6cb0' }}>Total Entries</span>
            <h2 style={{ fontSize: '32px', margin: '5px 0 0 0', color: '#2b6cb0' }}>{entrycount}</h2>
          </div>
          <div style={{ background: '#fff5f5', padding: '15px', borderRadius: '6px', textAlign: 'center' }}>
            <span style={{ fontSize: '12px', fontWeight: 'bold', color: '#c53030' }}>Total Exits</span>
            <h2 style={{ fontSize: '32px', margin: '5px 0 0 0', color: '#c53030' }}>{exitcount}</h2>
          </div>
        </div>

        <div style={{ background: '#f7fafc', padding: '12px', borderRadius: '6px', textAlign: 'center',
marginBottom: '20px', border: '1px dashed #cbd5e0' }}>
          <span style={{ fontSize: '13px', fontWeight: 'bold', color: '#4a5568' }}>Current People
Inside:</span>
          <span style={{ fontSize: '16px', fontWeight: 'bold', color: '#2d3748', marginLeft: '10px' }}>
            {currentPeople >= 0 ? currentPeople : 0}
          </span>
```

```
        </div>

        <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '15px' }}>
          <button
            onClick={this.updateEntry}
            style={{ padding: '10px', background: '#3182ce', color: '#fff', border: 'none', borderRadius: 
'6px', fontWeight: 'bold', cursor: 'pointer', fontSize: '14px', transition: 'background 0.2s' }}
          >
            Login (Enter)
          </button>
          <button
            onClick={this.updateExit}
            style={{ padding: '10px', background: '#e53e3e', color: '#fff', border: 'none', borderRadius: 
'6px', fontWeight: 'bold', cursor: 'pointer', fontSize: '14px', transition: 'background 0.2s' }}
          >
            Exit (Leave)
          </button>
        </div>
      </div>
    );
  }
}

export default function HOL8CounterApp() {
  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px 
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 8: Managing 
Component State</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Use class constructor and this.state to update counts via interactive triggers.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />
      <CountPeople />
    </div>
  );
}
```

HOL 9: Cricket Players ES6

Objective: Use ES6 features (map, filter, destructuring, spread, conditional render) to process cricket scores.

// File: HOL9CricketPlayers.jsx

```
import React, { useState } from 'react';

// 1. ListofPlayers Component
function ListofPlayers({ players }) {
  // Filter players with scores >= 70 (high scorers)
  const highScorers = players.filter(player => player.score >= 70);

  return (
    <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(240px, 1fr))', gap: '20px' }}>
      <div style={{ background: '#fff', border: '1px solid #e2e8f0', borderRadius: '8px', padding: '15px' }}>
        <h4 style={{ color: '#2b6cb0', margin: '0 0 10px 0', borderBottom: '2px solid #cbd5e0', paddingBottom:
'4px' }}>
          All Players List (Map)
        </h4>
        <ul style={{ listStyle: 'none', padding: 0, margin: 0 }}>
          {players.map((p, index) => (
            <li key={index} style={{ padding: '6px 0', borderBottom: '1px solid #edf2f7', fontSize: '13px',
display: 'flex', justifyContent: 'space-between' }}>
              <span>{index + 1}. {p.name}</span>
              <strong style={{ fontFamily: 'monospace' }}>{p.score} runs</strong>
            </li>
          ))}
        </ul>
      </div>

      <div style={{ background: '#fff', border: '1px solid #e2e8f0', borderRadius: '8px', padding: '15px' }}>
        <h4 style={{ color: '#276749', margin: '0 0 10px 0', borderBottom: '2px solid #c6f6d5', paddingBottom:
'4px' }}>
          Filtered Players (Score &ge; 70)
        </h4>
        <ul style={{ listStyle: 'none', padding: 0, margin: 0 }}>
          {highScorers.map((p, index) => (
            <li key={index} style={{ padding: '6px 0', borderBottom: '1px solid #edf2f7', fontSize: '13px',
display: 'flex', justifyContent: 'space-between', color: '#22543d' }}>
              <span>? {p.name}</span>
              <strong style={{ fontFamily: 'monospace' }}>{p.score} runs</strong>
            </li>
          ))}
        </ul>
      </div>
    </div>
  );
}

// 2. IndianPlayers Component
function IndianPlayers() {
  const allIndianPlayers = ['Rohit Sharma', 'Virat Kohli', 'KL Rahul', 'Shreyas Iyer', 'Rishabh Pant', 'Hardik
Pandya', 'Ravindra Jadeja', 'Jasprit Bumrah', 'Mohammed Shami', 'Kuldeep Yadav', 'Mohammed Siraj'];

  // Destructuring Odd vs Even position players (using 0-indexed values: index 0 is first, index 1 is second,
etc.)
  // Let's destructure the first 4 elements
  const [first, second, third, fourth, ...rest] = allIndianPlayers;
  const oddTeam = [first, third, rest[0], rest[2], rest[4]];   // Positions 1, 3, 5, 7, 9
  const evenTeam = [second, fourth, rest[1], rest[3], rest[5]]; // Positions 2, 4, 6, 8, 10
```

```
  // Merge T20players and RanjiTrophy players using spread operator
  const t20Players = ['Suryakumar Yadav', 'Yashasvi Jaiswal', 'Rinku Singh'];
  const ranjiPlayers = ['Sarfaraz Khan', 'Abhimanyu Easwaran', 'Baba Indrajith'];
  const mergedPlayers = [...t20Players, ...ranjiPlayers];

  return (
    <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(220px, 1fr))', gap: '20px' }}>
      <div style={{ background: '#fff', border: '1px solid #e2e8f0', borderRadius: '8px', padding: '15px' }}>
        <h4 style={{ color: '#805ad5', margin: '0 0 10px 0', borderBottom: '2px solid #e9d8fd', paddingBottom:
'4px' }}>
          Destructured Teams
        </h4>
        <div style={{ fontSize: '13px' }}>
          <strong style={{ color: '#553c9a' }}>Odd Team Players:</strong>
          <ul style={{ paddingLeft: '20px', margin: '5px 0 10px 0' }}>
            {oddTeam.map((p, idx) => <li key={idx}>{p}</li>)}
          </ul>
          <strong style={{ color: '#553c9a' }}>Even Team Players:</strong>
          <ul style={{ paddingLeft: '20px', margin: '5px 0 0 0' }}>
            {evenTeam.map((p, idx) => <li key={idx}>{p}</li>)}
          </ul>
        </div>
      </div>

      <div style={{ background: '#fff', border: '1px solid #e2e8f0', borderRadius: '8px', padding: '15px' }}>
        <h4 style={{ color: '#dd6b20', margin: '0 0 10px 0', borderBottom: '2px solid #ffe8cc', paddingBottom:
'4px' }}>
          Merged Squad (Spread)
        </h4>
        <p style={{ fontSize: '12px', color: '#718096', margin: '0 0 10px 0' }}>
          T20 Squad + Ranji Trophy Squad merged:
        </p>
        <ul style={{ listStyle: 'none', padding: 0, margin: 0 }}>
          {mergedPlayers.map((p, index) => (
            <li key={index} style={{ padding: '6px 0', borderBottom: '1px solid #edf2f7', fontSize: '13px',
display: 'flex', alignItems: 'center' }}>
              <span style={{ display: 'inline-block', width: '8px', height: '8px', borderRadius: '50%',
background: '#dd6b20', marginRight: '8px' }}></span>
              {p}
            </li>
          ))}
        </ul>
      </div>
    </div>
  );
}

// 3. Main HOL 9 Component with boolean Flag toggle
export default function HOL9CricketPlayers() {
  const [flag, setFlag] = useState(true);

  // 11 Mock Players with name and score
  const mockPlayers = [
    { name: 'Sachin Tendulkar', score: 100 },
    { name: 'Virat Kohli', score: 85 },
    { name: 'MS Dhoni', score: 55 },
    { name: 'Rohit Sharma', score: 92 },
    { name: 'Yuvraj Singh', score: 64 },
    { name: 'Suresh Raina', score: 45 },
    { name: 'Gautam Gambhir', score: 75 },
    { name: 'Hardik Pandya', score: 38 },
```

```
    { name: 'Ravindra Jadeja', score: 72 },
    { name: 'Shikhar Dhawan', score: 68 },
    { name: 'Rishabh Pant', score: 81 }
  ];

  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 9: ES6 & Functional
Programming</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Use ES6 map, filter, array/object destructuring, spread operators, and conditional flags to
display data.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />

      <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom:
'20px', padding: '10px 15px', background: '#f7fafc', borderRadius: '6px', border: '1px solid #e2e8f0' }}>
        <span style={{ fontSize: '13px', fontWeight: 'bold', color: '#4a5568' }}>
          Current Display Flag: <code style={{ color: '#3182ce' }}>flag = {flag.toString()}</code>
        </span>
        <button onClick={() => setFlag(!flag)} style={{ background: '#3182ce', color: '#fff', border: 'none',
padding: '6px 12px', borderRadius: '4px', cursor: 'pointer', fontSize: '12px', fontWeight: 'bold' }}>
          Toggle Flag View
        </button>
      </div>

      {/* Conditional rendering based on flag */}
      {flag ? (
        <div>
          <h3 style={{ color: '#2d3748', fontSize: '16px', marginBottom: '15px' }}>Showing Component:
ListofPlayers (Flag is True)</h3>
          <ListofPlayers players={mockPlayers} />
        </div>
      ) : (
        <div>
          <h3 style={{ color: '#2d3748', fontSize: '16px', marginBottom: '15px' }}>Showing Component:
IndianPlayers (Flag is False)</h3>
          <IndianPlayers />
        </div>
      )}
    </div>
  );
}
```

HOL 10: Office Space JSX

Objective: Use JSX attributes to render office rentals and color rent status dynamically (Red < 60000, Green otherwise).

// File: HOL10OfficeSpace.jsx

```
import React from 'react';

export default function HOL10OfficeSpace() {
  // Mock office space details
  const offices = [
    { id: 1, name: 'Premium Tech Hub', rent: 75000, address: 'Sector 62, Noida, UP', image:
'https://images.unsplash.com/photo-1497366216548-37526070297c?auto=format&fit=crop&w=400&h=250&q=80' },
    { id: 2, name: 'Co-Working Startup Nest', rent: 48000, address: 'MG Road, Bengaluru, KA', image:
'https://images.unsplash.com/photo-1524758631624-e2822e304c36?auto=format&fit=crop&w=400&h=250&q=80' },
    { id: 3, name: 'Enterprise Corporate Office', rent: 125000, address: 'Cyber City, Gurugram, HR', image:
'https://images.unsplash.com/photo-1497215728101-856f4ea42174?auto=format&fit=crop&w=400&h=250&q=80' },
    { id: 4, name: 'Creative Studio Cabin', rent: 55000, address: 'Andheri West, Mumbai, MH', image:
'https://images.unsplash.com/photo-1531973576160-7125cd663d86?auto=format&fit=crop&w=400&h=250&q=80' },
    { id: 5, name: 'Executive Suite', rent: 90000, address: 'Salt Lake Sector V, Kolkata, WB', image:
'https://images.unsplash.com/photo-1497366811353-6870744d04b2?auto=format&fit=crop&w=400&h=250&q=80' }
  ];

  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 id="office-rental-heading" style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>
        Hands-on Lab 10: JSX Attributes & Conditional CSS Styling
      </h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Loop through elements in JSX, display image attributes, and style rents
(<strong>Red</strong> if &lt; 60,000, <strong>Green</strong> if &ge; 60,000).
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />

      <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(280px, 1fr))', gap: '20px',
marginTop: '10px' }}>
        {offices.map(office => {
          // Dynamic Rent text color styling
          const rentColor = office.rent < 60000 ? '#e53e3e' : '#38a169';

          return (
            <div key={office.id} style={{ border: '1px solid #e2e8f0', borderRadius: '8px', overflow: 'hidden',
background: '#fff', boxShadow: '0 2px 4px rgba(0,0,0,0.02)' }}>
              {/* Image with attributes */}
              <img
                src={office.image}
                alt={office.name}
                style={{ width: '100%', height: '160px', objectFit: 'cover', display: 'block' }}
              />
              <div style={{ padding: '15px' }}>
                <h4 style={{ margin: '0 0 8px 0', fontSize: '16px', color: '#2d3748' }}>{office.name}</h4>
                <p style={{ margin: '0 0 10px 0', fontSize: '13px', color: '#718096' }}>
                  <strong>Location:  Address:</strong> {office.address}
                </p>
                <div style={{ borderTop: '1px solid #edf2f7', paddingTop: '10px', display: 'flex',
justifyContent: 'space-between', alignItems: 'center' }}>
                  <span style={{ fontSize: '12px', fontWeight: 'bold', color: '#4a5568' }}>Monthly Rent:</span>
                  <span style={{ fontSize: '16px', fontWeight: 'bold', color: rentColor, fontFamily:
'monospace' }}>
                    ?{office.rent.toLocaleString()}/-
```

```
                  </span>
                </div>
              </div>
            </div>
          );
        })}
      </div>
    </div>
  );
}
```

HOL 11: Event Examples

Objective: Trigger multi-method updates, pass arguments to handlers, and log synthetic events.

// File: HOL11EventExamples.jsx

```
import React, { useState } from 'react';

export default function HOL11EventExamples() {
  const [counter, setCounter] = useState(0);
  const [logs, setLogs] = useState([]);

  const addLog = (msg) => {
    setLogs(prev => [msg, ...prev.slice(0, 9)]);
  };

  // 1. Multi-method handlers for Increment
  const handleIncrement = (e) => {
    // Method 1: Increment state counter
    setCounter(prev => prev + 1);

    // Method 2: Log / Alert action
    addLog(`Incremented counter. Current temp value: ${counter + 1}`);
  };

  const handleDecrement = () => {
    setCounter(prev => prev - 1);
    addLog(`Decremented counter. Current temp value: ${counter - 1}`);
  };

  // 2. Passing arguments to function
  const handleSayWelcome = (message) => {
    alert(`Function argument received: "${message}"`);
    addLog(`Invoked welcome function with argument: "${message}"`);
  };

  // 3. Synthetic Event Handler
  const handleSyntheticEvent = (event) => {
    alert(`I was clicked! Synthetic Event Type: "${event.type}"`);
    addLog(`Synthetic Event captured: click on button "${event.target.innerText}"`);
  };

  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 11: Handling React
Events</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Invoke multiple methods on click, pass parameters in handlers, and intercept Synthetic
Events.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />

      <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(280px, 1fr))', gap: '20px'
}}>

        {/* Actions panel */}
        <div style={{ display: 'flex', flexDirection: 'column', gap: '15px' }}>

          {/* Counter section */}
          <div style={{ background: '#f7fafc', padding: '15px', borderRadius: '6px', border: '1px solid
```

```
#e2e8f0' }}>
            <h4 style={{ margin: '0 0 10px 0', color: '#2d3748' }}>1. Counter (Multi-Method Click)</h4>
            <div style={{ display: 'flex', alignItems: 'center', gap: '15px' }}>
              <button onClick={handleDecrement} style={{ padding: '6px 12px', background: '#e53e3e', color:
'#fff', border: 'none', borderRadius: '4px', cursor: 'pointer', fontWeight: 'bold' }}>
                - Decrement
              </button>
              <span style={{ fontSize: '20px', fontWeight: 'bold', minWidth: '40px', textAlign: 'center'
}}>{counter}</span>
              <button onClick={handleIncrement} style={{ padding: '6px 12px', background: '#38a169', color:
'#fff', border: 'none', borderRadius: '4px', cursor: 'pointer', fontWeight: 'bold' }}>
                + Increment
              </button>
            </div>
            <p style={{ margin: '8px 0 0 0', fontSize: '11px', color: '#718096' }}>
              * Increment invokes state change and appends to the logger list below.
            </p>
          </div>

          {/* Argument passing section */}
          <div style={{ background: '#f7fafc', padding: '15px', borderRadius: '6px', border: '1px solid
#e2e8f0' }}>
            <h4 style={{ margin: '0 0 10px 0', color: '#2d3748' }}>2. Pass Arguments to Handler</h4>
            <button
              onClick={() => handleSayWelcome('welcome')}
              style={{ padding: '8px 14px', background: '#3182ce', color: '#fff', border: 'none', borderRadius:
'4px', cursor: 'pointer', fontWeight: 'bold', fontSize: '13px' }}
            >
              Say Welcome (Pass "welcome")
            </button>
          </div>

          {/* Synthetic Event section */}
          <div style={{ background: '#f7fafc', padding: '15px', borderRadius: '6px', border: '1px solid
#e2e8f0' }}>
            <h4 style={{ margin: '0 0 10px 0', color: '#2d3748' }}>3. Synthetic Event click</h4>
            <button
              onClick={handleSyntheticEvent}
              style={{ padding: '8px 14px', background: '#805ad5', color: '#fff', border: 'none', borderRadius:
'4px', cursor: 'pointer', fontWeight: 'bold', fontSize: '13px' }}
            >
              Trigger Synthetic Event
            </button>
          </div>

        </div>

        {/* Logger panel */}
        <div style={{ background: '#2d3748', color: '#a0aec0', padding: '15px', borderRadius: '6px',
fontFamily: 'monospace', fontSize: '12px' }}>
          <h4 style={{ color: '#fff', margin: '0 0 10px 0', borderBottom: '1px solid #4a5568', paddingBottom:
'5px' }}>Event Action Logger</h4>
          {logs.length === 0 ? (
            <div style={{ color: '#718096', fontStyle: 'italic' }}>No events triggered yet...</div>
          ) : (
            <ul style={{ listStyle: 'none', padding: 0, margin: 0 }}>
              {logs.map((log, index) => (
                <li key={index} style={{ marginBottom: '6px', borderBottom: '1px solid #4a5568', paddingBottom:
'3px', color: '#cbd5e0' }}>
                  &gt; {log}
                </li>
              ))}
```

```
            </ul>
          )}
        </div>

      </div>
    </div>
  );
}
```

HOL 12: Conditional Login

Objective: Switch views based on login/logout state.

// File: HOL12LoginLogout.jsx

```
import React, { useState } from 'react';

// User Page Component
function UserPage({ onLogout }) {
  return (
    <div style={{ padding: '20px', background: '#e6fffa', border: '1px solid #b2f5ea', borderRadius: '8px',
textAlign: 'center' }}>
      <h3 style={{ color: '#234e52', margin: '0 0 10px 0' }}>Welcome back, Authorized User!</h3>
      <p style={{ fontSize: '14px', color: '#2d3748', marginBottom: '15px' }}>
        You have successfully authenticated. You now have access to your personal portal dashboard and account
settings.
      </p>
      <button
        onClick={onLogout}
        style={{ padding: '8px 16px', background: '#319795', color: '#fff', border: 'none', borderRadius:
'4px', cursor: 'pointer', fontWeight: 'bold' }}
      >
        Sign Out / Logout
      </button>
    </div>
  );
}

// Guest Page Component
function GuestPage({ onLogin }) {
  return (
    <div style={{ padding: '20px', background: '#fffaf0', border: '1px solid #feebc8', borderRadius: '8px',
textAlign: 'center' }}>
      <h3 style={{ color: '#7b341e', margin: '0 0 10px 0' }}>Hello Guest!</h3>
      <p style={{ fontSize: '14px', color: '#2d3748', marginBottom: '15px' }}>
        Please log in to your account to view secure content and access student management records.
      </p>
      <button
        onClick={onLogin}
        style={{ padding: '8px 16px', background: '#dd6b20', color: '#fff', border: 'none', borderRadius:
'4px', cursor: 'pointer', fontWeight: 'bold' }}
      >
        Sign In / Login
      </button>
    </div>
  );
}

export default function HOL12LoginLogout() {
  const [isLoggedIn, setIsLoggedIn] = useState(false);

  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 12: Conditional
Rendering (User vs Guest)</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Dynamically toggle page templates depending on authentication state.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />
```

```
      <div style={{ maxWidth: '400px', margin: '20px auto' }}>
        {isLoggedIn ? (
          <UserPage onLogout={() => setIsLoggedIn(false)} />
        ) : (
          <GuestPage onLogin={() => setIsLoggedIn(true)} />
        )}
      </div>
    </div>
  );
}
```

HOL 13: Blogger App

Objective: Demonstrate multiple conditional rendering methods (ternary, AND, element variables).

// File: HOL13BloggerApp.jsx

```
import React, { useState } from 'react';

// Sub-components
function BookDetails() {
  return (
    <div style={{ padding: '15px', background: '#ebf8ff', borderRadius: '6px', border: '1px solid #bee3f8' }}>
      <h4 style={{ color: '#2b6cb0', margin: '0 0 8px 0' }}>? Book Details</h4>
      <p style={{ fontSize: '13px', margin: '0 0 5px 0' }}><strong>Title:</strong> React Design Patterns & Best
Practices</p>
      <p style={{ fontSize: '13px', margin: '0 0 5px 0' }}><strong>Author:</strong> Michele Bertoli</p>
      <p style={{ fontSize: '13px', margin: 0 }}><strong>Description:</strong> Build modular, clean, and
high-performance React web applications.</p>
    </div>
  );
}

function BlogDetails() {
  return (
    <div style={{ padding: '15px', background: '#f0fff4', borderRadius: '6px', border: '1px solid #c6f6d5' }}>
      <h4 style={{ color: '#276749', margin: '0 0 8px 0' }}>? Blog Details</h4>
      <p style={{ fontSize: '13px', margin: '0 0 5px 0' }}><strong>Title:</strong> Mastering React State: Redux
vs Context API</p>
      <p style={{ fontSize: '13px', margin: '0 0 5px 0' }}><strong>Author:</strong> Dan Abramov</p>
      <p style={{ fontSize: '13px', margin: 0 }}><strong>Published:</strong> July 10, 2026 ? 8 min read</p>
    </div>
  );
}

function CourseDetails() {
  return (
    <div style={{ padding: '15px', background: '#fffaf0', borderRadius: '6px', border: '1px solid #feebc8' }}>
      <h4 style={{ color: '#dd6b20', margin: '0 0 8px 0' }}>? Course Details</h4>
      <p style={{ fontSize: '13px', margin: '0 0 5px 0' }}><strong>Title:</strong> Full-Stack Development with
Spring Boot & React</p>
      <p style={{ fontSize: '13px', margin: '0 0 5px 0' }}><strong>Provider:</strong> Cognizant Academy</p>
      <p style={{ fontSize: '13px', margin: 0 }}><strong>Duration:</strong> 12 Weeks ? Self-paced deep
skilling</p>
    </div>
  );
}

export default function HOL13BloggerApp() {
  const [selectedTab, setSelectedTab] = useState('book');

  // Conditional Rendering Method 1: Element Variables
  let renderedDetailsElement;
  if (selectedTab === 'book') {
    renderedDetailsElement = <BookDetails />;
  } else if (selectedTab === 'blog') {
    renderedDetailsElement = <BlogDetails />;
  } else if (selectedTab === 'course') {
    renderedDetailsElement = <CourseDetails />;
  }

  return (
```

```
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 13: Advanced
Conditional Rendering</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Render Book, Blog, and Course components using multiple conditional rendering patterns
(if-else element variables, ternary operator, logical AND).
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />

      {/* Tab selectors */}
      <div style={{ display: 'flex', gap: '10px', marginBottom: '20px' }}>
        <button
          onClick={() => setSelectedTab('book')}
          style={{ padding: '8px 16px', background: selectedTab === 'book' ? '#3182ce' : '#edf2f7', color:
selectedTab === 'book' ? '#fff' : '#4a5568', border: 'none', borderRadius: '6px', cursor: 'pointer',
fontWeight: 'bold', fontSize: '13px' }}
        >
          Book Details
        </button>
        <button
          onClick={() => setSelectedTab('blog')}
          style={{ padding: '8px 16px', background: selectedTab === 'blog' ? '#38a169' : '#edf2f7', color:
selectedTab === 'blog' ? '#fff' : '#4a5568', border: 'none', borderRadius: '6px', cursor: 'pointer',
fontWeight: 'bold', fontSize: '13px' }}
        >
          Blog Details
        </button>
        <button
          onClick={() => setSelectedTab('course')}
          style={{ padding: '8px 16px', background: selectedTab === 'course' ? '#dd6b20' : '#edf2f7', color:
selectedTab === 'course' ? '#fff' : '#4a5568', border: 'none', borderRadius: '6px', cursor: 'pointer',
fontWeight: 'bold', fontSize: '13px' }}
        >
          Course Details
        </button>
      </div>

      <div style={{ display: 'flex', flexDirection: 'column', gap: '20px' }}>
        {/* Style 1 Demonstration */}
        <div style={{ border: '1px dashed #cbd5e0', padding: '15px', borderRadius: '6px' }}>
          <h5 style={{ margin: '0 0 10px 0', color: '#4a5568', fontSize: '11px', textTransform: 'uppercase' }}>
            Pattern 1: Render via Element Variable
          </h5>
          {renderedDetailsElement}
        </div>

        {/* Style 2 Demonstration */}
        <div style={{ border: '1px dashed #cbd5e0', padding: '15px', borderRadius: '6px' }}>
          <h5 style={{ margin: '0 0 10px 0', color: '#4a5568', fontSize: '11px', textTransform: 'uppercase' }}>
            Pattern 2: Render via Ternary Operator (Inline)
          </h5>
          {selectedTab === 'book' ? (
            <BookDetails />
          ) : selectedTab === 'blog' ? (
            <BlogDetails />
          ) : (
            <CourseDetails />
          )}
        </div>

        {/* Style 3 Demonstration */}
```

```
        <div style={{ border: '1px dashed #cbd5e0', padding: '15px', borderRadius: '6px' }}>
          <h5 style={{ margin: '0 0 10px 0', color: '#4a5568', fontSize: '11px', textTransform: 'uppercase' }}>
            Pattern 3: Render via Logical AND (&&) Operators
          </h5>
          {selectedTab === 'book' && <BookDetails />}
          {selectedTab === 'blog' && <BlogDetails />}
          {selectedTab === 'course' && <CourseDetails />}
        </div>
      </div>
    </div>
  );
}
```

HOL 14: Context API Dynamic Theme

Objective: Implement a theme toggler bypassing props using React Context Provider.

// File: HOL14ContextAPI.jsx

```
import React, { useState, useContext } from 'react';
import ThemeContext from './ThemeContext';

// 1. EmployeeCard Child Component (Consumes ThemeContext)
function EmployeeCard({ employee }) {
  const theme = useContext(ThemeContext);

  // Define styling dynamically based on context theme value
  const cardStyle = theme === 'dark'
    ? { background: '#2d3748', color: '#f7fafc', border: '1px solid #4a5568' }
    : { background: '#fff', color: '#2d3748', border: '1px solid #e2e8f0' };

  const roleStyle = theme === 'dark' ? '#cbd5e0' : '#718096';

  return (
    <div style={{ padding: '15px', borderRadius: '6px', boxShadow: '0 2px 4px rgba(0,0,0,0.02)', transition:
'all 0.2s', ...cardStyle }}>
      <h4 style={{ margin: '0 0 5px 0', fontSize: '15px' }}>{employee.name}</h4>
      <p style={{ margin: '0 0 8px 0', fontSize: '12px', color: roleStyle, fontWeight: '600' }}>
        Role: {employee.role}
      </p>
      <p style={{ margin: 0, fontSize: '12px', fontStyle: 'italic' }}>
        Email: {employee.email}
      </p>
    </div>
  );
}

// 2. EmployeesList Component
function EmployeesList({ employees }) {
  return (
    <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(200px, 1fr))', gap: '15px' }}>
      {employees.map(emp => (
        <EmployeeCard key={emp.id} employee={emp} />
      ))}
    </div>
  );
}

// 3. Main HOL 14 Component
export default function HOL14ContextAPI() {
  const [theme, setTheme] = useState('light');

  const mockEmployees = [
    { id: 1, name: 'Siddharth Sen', role: 'Software Engineer', email: 'siddharth@cognizant.com' },
    { id: 2, name: 'Neha Sharma', role: 'UX Designer', email: 'neha@cognizant.com' },
    { id: 3, name: 'David Jones', role: 'DevOps Architect', email: 'david@cognizant.com' }
  ];

  const handleToggleTheme = () => {
    setTheme(prev => prev === 'light' ? 'dark' : 'light');
  };

  const containerBg = theme === 'dark' ? '#1a202c' : '#f7fafc';
  const labelStyle = theme === 'dark' ? '#cbd5e0' : '#4a5568';
```

```
  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 14: React Context API</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Replace prop drilling by wrapping components in a Context Provider and accessing values using useContext.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />

      {/* Controller to toggle Context Value */}
      <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: '20px', padding: '10px 15px', background: '#edf2f7', borderRadius: '6px' }}>
        <span style={{ fontSize: '13px', fontWeight: 'bold', color: '#4a5568' }}>
          Active Theme Context: <code style={{ color: '#3182ce', textTransform: 'uppercase' }}>{theme}</code>
        </span>
        <button
          onClick={handleToggleTheme}
          style={{ background: '#2d3748', color: '#fff', border: 'none', padding: '6px 12px', borderRadius: '4px', cursor: 'pointer', fontSize: '12px', fontWeight: 'bold' }}
        >
          Toggle Theme Context ({theme === 'light' ? 'Dark' : 'Light'})
        </button>
      </div>

      {/* The Provider wrapping children */}
      <ThemeContext.Provider value={theme}>
        <div style={{ padding: '20px', background: containerBg, borderRadius: '8px', transition: 'background-color 0.2s' }}>
          <h3 style={{ margin: '0 0 15px 0', fontSize: '16px', color: theme === 'dark' ? '#fff' : '#2d3748' }}>
            Employees Directory
          </h3>
          <EmployeesList employees={mockEmployees} />
        </div>
      </ThemeContext.Provider>
    </div>
  );
}
```

HOL 15: Ticket Register

Objective: Form validation controlled component with reference ID alert on submit.

// File: HOL15TicketRegister.jsx

```
import React, { useState } from 'react';

// The ComplaintRegister form component
export function ComplaintRegister() {
  const [employeeName, setEmployeeName] = useState('');
  const [complaintText, setComplaintText] = useState('');
  const [tickets, setTickets] = useState([]);

  const handleSubmit = (e) => {
    e.preventDefault();

    if (!employeeName.trim() || !complaintText.trim()) {
      alert('Please fill out both the employee name and complaint fields.');
      return;
    }

    // Generate a random Reference Number (e.g. REF-17382-2026)
    const refNum = `REF-${Math.floor(10000 + Math.random() * 90000)}-${new Date().getFullYear()}`;

    alert(`Complaint Submitted successfully!\n\nReference Number: ${refNum}\nEmployee: ${employeeName}`);

    // Add to local list for visual verification
    const newTicket = {
      id: Date.now(),
      ref: refNum,
      name: employeeName,
      complaint: complaintText,
      date: new Date().toLocaleDateString()
    };
    setTickets(prev => [newTicket, ...prev]);

    // Reset form fields
    setEmployeeName('');
    setComplaintText('');
  };

  return (
    <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(280px, 1fr))', gap: '20px' }}>

      {/* Raising form */}
      <form onSubmit={handleSubmit} style={{ background: '#f7fafc', padding: '20px', borderRadius: '8px',
border: '1px solid #e2e8f0' }}>
        <h3 style={{ color: '#2b6cb0', margin: '0 0 15px 0', fontSize: '16px' }}>Register New Complaint</h3>

        <div style={{ marginBottom: '15px' }}>
          <label style={{ display: 'block', fontSize: '13px', fontWeight: 'bold', color: '#4a5568',
marginBottom: '5px' }}>
            Employee Name:
          </label>
          <input
            type="text"
            value={employeeName}
            onChange={(e) => setEmployeeName(e.target.value)}
            placeholder="Enter employee name"
            style={{ width: '100%', padding: '8px', border: '1px solid #cbd5e0', borderRadius: '4px', fontSize:
```

```
'13px' }}
          />
        </div>

        <div style={{ marginBottom: '15px' }}>
          <label style={{ display: 'block', fontSize: '13px', fontWeight: 'bold', color: '#4a5568',
marginBottom: '5px' }}>
            Complaint Description:
          </label>
          <textarea
            rows="4"
            value={complaintText}
            onChange={(e) => setComplaintText(e.target.value)}
            placeholder="Type complaint details..."
            style={{ width: '100%', padding: '8px', border: '1px solid #cbd5e0', borderRadius: '4px', fontSize:
'13px', resize: 'vertical' }}
          />
        </div>

        <button
          type="submit"
          style={{ width: '100%', padding: '10px', background: '#3182ce', color: '#fff', border: 'none',
borderRadius: '4px', fontWeight: 'bold', cursor: 'pointer', fontSize: '14px' }}
        >
          Submit Complaint
        </button>
      </form>

      {/* Submitted logs list */}
      <div style={{ background: '#fff', border: '1px solid #e2e8f0', borderRadius: '8px', padding: '20px' }}>
        <h3 style={{ color: '#4a5568', margin: '0 0 15px 0', fontSize: '16px' }}>Active Tickets / Complaint
History</h3>
        {tickets.length === 0 ? (
          <div style={{ color: '#a0aec0', fontStyle: 'italic', textAlign: 'center', marginTop: '30px' }}>
            No complaints raised yet. Use the form to raise a complaint.
          </div>
        ) : (
          <div style={{ display: 'flex', flexDirection: 'column', gap: '10px', maxHeight: '300px', overflowY:
'auto' }}>
            {tickets.map(t => (
              <div key={t.id} style={{ border: '1px solid #edf2f7', borderRadius: '6px', padding: '12px',
background: '#fffaf0' }}>
                <div style={{ display: 'flex', justifyContent: 'space-between', fontSize: '12px', marginBottom:
'5px' }}>
                  <span style={{ fontWeight: 'bold', color: '#b7791f' }}>{t.ref}</span>
                  <span style={{ color: '#718096' }}>{t.date}</span>
                </div>
                <strong style={{ fontSize: '13px', color: '#2d3748' }}>Employee: {t.name}</strong>
                <p style={{ margin: '5px 0 0 0', fontSize: '12px', color: '#4a5568', lineHeight: '1.4' }}>
                  {t.complaint}
                </p>
              </div>
            ))}
          </div>
        )}
      </div>

    </div>
  );
}

export default function HOL15TicketRegister() {
```

```
  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 15: Forms &
Alerts</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Implement controlled form elements, handle form submissions, and generate reference tracking
alerts.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />
      <ComplaintRegister />
    </div>
  );
}
```

HOL 16: Mail Register Validation

Objective: Complete email, name length, password validation rules with form errors display.

// File: HOL16MailRegister.jsx

```
import React, { useState } from 'react';

export function Register() {
  const [formData, setFormData] = useState({ name: '', email: '', password: '' });
  const [errors, setErrors] = useState({});
  const [isSubmitted, setIsSubmitted] = useState(false);

  // Validate fields
  const validateField = (name, value) => {
    let errorMsg = '';
    if (name === 'name') {
      if (value.trim().length < 5) {
        errorMsg = 'Name must be at least 5 characters long.';
      }
    } else if (name === 'email') {
      if (!value.includes('@') || !value.includes('.')) {
        errorMsg = 'Email must contain "@" and "." (e.g. user@example.com).';
      }
    } else if (name === 'password') {
      if (value.length < 8) {
        errorMsg = 'Password must be at least 8 characters long.';
      }
    }
    return errorMsg;
  };

  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData(prev => ({ ...prev, [name]: value }));

    // Validate on change
    const errorMsg = validateField(name, value);
    setErrors(prev => ({ ...prev, [name]: errorMsg }));
    setIsSubmitted(false); // Reset success alert if they edit the form
  };

  const handleSubmit = (e) => {
    e.preventDefault();

    // Validate all fields on submit
    const newErrors = {};
    Object.keys(formData).forEach(key => {
      const errorMsg = validateField(key, formData[key]);
      if (errorMsg) {
        newErrors[key] = errorMsg;
      }
    });

    setErrors(newErrors);

    // If no errors, registration is successful
    if (Object.keys(newErrors).length === 0) {
      alert(`Registration Successful!\n\nWelcome: ${formData.name}\nEmail: ${formData.email}`);
      setIsSubmitted(true);
      // Reset form
```

```
      setFormData({ name: '', email: '', password: '' });
      setErrors({});
    } else {
      alert('Registration Failed! Please fix the errors in the form.');
    }
  };

  return (
    <div style={{ maxWidth: '400px', margin: '0 auto', background: '#fff', padding: '20px', borderRadius:
'8px', border: '1px solid #e2e8f0', boxShadow: '0 4px 6px rgba(0,0,0,0.02)' }}>
      <h3 style={{ color: '#2b6cb0', margin: '0 0 15px 0', textAlign: 'center' }}>Mail Account
Registration</h3>

      <form onSubmit={handleSubmit}>

        {/* Name Field */}
        <div style={{ marginBottom: '12px' }}>
          <label style={{ display: 'block', fontSize: '13px', fontWeight: 'bold', color: '#4a5568',
marginBottom: '4px' }}>Name</label>
          <input
            type="text"
            name="name"
            value={formData.name}
            onChange={handleChange}
            placeholder="Min 5 characters"
            style={{ width: '100%', padding: '8px', border: errors.name ? '1px solid #e53e3e' : '1px solid
#cbd5e0', borderRadius: '4px', fontSize: '13px' }}
          />
          {errors.name && <span style={{ fontSize: '11px', color: '#e53e3e', display: 'block', marginTop: '3px'
}}>{errors.name}</span>}
        </div>

        {/* Email Field */}
        <div style={{ marginBottom: '12px' }}>
          <label style={{ display: 'block', fontSize: '13px', fontWeight: 'bold', color: '#4a5568',
marginBottom: '4px' }}>Email Address</label>
          <input
            type="text"
            name="email"
            value={formData.email}
            onChange={handleChange}
            placeholder="user@domain.com"
            style={{ width: '100%', padding: '8px', border: errors.email ? '1px solid #e53e3e' : '1px solid
#cbd5e0', borderRadius: '4px', fontSize: '13px' }}
          />
          {errors.email && <span style={{ fontSize: '11px', color: '#e53e3e', display: 'block', marginTop:
'3px' }}>{errors.email}</span>}
        </div>

        {/* Password Field */}
        <div style={{ marginBottom: '20px' }}>
          <label style={{ display: 'block', fontSize: '13px', fontWeight: 'bold', color: '#4a5568',
marginBottom: '4px' }}>Password</label>
          <input
            type="password"
            name="password"
            value={formData.password}
            onChange={handleChange}
            placeholder="Min 8 characters"
            style={{ width: '100%', padding: '8px', border: errors.password ? '1px solid #e53e3e' : '1px solid
#cbd5e0', borderRadius: '4px', fontSize: '13px' }}
          />
```

```
          {errors.password && <span style={{ fontSize: '11px', color: '#e53e3e', display: 'block', marginTop:
'3px' }}>{errors.password}</span>}
        </div>

        <button
          type="submit"
          style={{ width: '100%', padding: '10px', background: '#3182ce', color: '#fff', border: 'none',
borderRadius: '4px', fontWeight: 'bold', cursor: 'pointer', fontSize: '14px' }}
        >
          Register Mail Account
        </button>

      </form>

      {isSubmitted && (
        <div style={{ marginTop: '15px', padding: '10px', background: '#c6f6d5', color: '#22543d', border: '1px
solid #9ae6b4', borderRadius: '6px', fontSize: '12px', textAlign: 'center', fontWeight: 'bold' }}>
          [MET] Mail Registration Verified & Recorded!
        </div>
      )}
    </div>
  );
}

export default function HOL16MailRegister() {
  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 16: Form
Validation</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Perform field-level validation (name &ge; 5 chars, email contains @ and ., password &ge; 8
chars) on key inputs.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />
      <Register />
    </div>
  );
}
```

HOL 17: FetchUser RandomUser Async

Objective: Asynchronously fetch random user profiles using Fetch API inside componentDidMount.

// File: HOL17FetchUser.jsx

```
import React from 'react';

// The Getuser class-based component
export class Getuser extends React.Component {
  constructor(props) {
    super(props);
    this.state = {
      userData: null,
      loading: true,
      error: null
    };
  }

  fetchUser() {
    this.setState({ loading: true, error: null });
    fetch('https://api.randomuser.me/')
      .then(response => {
        if (!response.ok) {
          throw new Error('Failed to fetch user data');
        }
        return response.json();
      })
      .then(data => {
        const user = data.results[0];
        this.setState({
          userData: {
            title: user.name.title,
            first: user.name.first,
            last: user.name.last,
            image: user.picture.large,
            email: user.email,
            location: `${user.location.city}, ${user.location.country}`
          },
          loading: false
        });
      })
      .catch(error => {
        this.setState({ error: error.message, loading: false });
      });
  }

  componentDidMount() {
    this.fetchUser();
  }

  render() {
    const { userData, loading, error } = this.state;

    if (loading) {
      return (
        <div style={{ textAlign: 'center', padding: '30px', color: '#4a5568', fontWeight: 'bold' }}>
          Fetching random user profile...
        </div>
      );
    }
```

```
    if (error) {
      return (
        <div style={{ color: '#e53e3e', padding: '15px', background: '#fff5f5', borderRadius: '6px', border:
'1px solid #fed7d7', textAlign: 'center' }}>
          Error: {error}. Please try again.
        </div>
      );
    }

    return (
      <div style={{ maxWidth: '350px', margin: '0 auto', background: '#fff', border: '1px solid #e2e8f0',
borderRadius: '8px', padding: '20px', boxShadow: '0 4px 6px rgba(0,0,0,0.02)', textAlign: 'center' }}>
        <img
          src={userData.image}
          alt={userData.first}
          style={{ width: '120px', height: '120px', borderRadius: '50%', objectFit: 'cover', margin: '0 auto
15px auto', border: '3px solid #3182ce', display: 'block' }}
        />

        <h3 style={{ margin: '0 0 5px 0', fontSize: '18px', color: '#2d3748', textTransform: 'capitalize' }}>
          {userData.title}. {userData.first} {userData.last}
        </h3>

        <p style={{ fontSize: '12px', color: '#718096', margin: '0 0 15px 0' }}>
          Email:  {userData.email} <br />
          Location:  {userData.location}
        </p>

        <button
          onClick={() => this.fetchUser()}
          style={{ width: '100%', padding: '8px 12px', background: '#3182ce', color: '#fff', border: 'none',
borderRadius: '4px', cursor: 'pointer', fontWeight: 'bold', fontSize: '13px' }}
        >
          Load Next Random User
        </button>
      </div>
    );
  }
}

export default function HOL17FetchUser() {
  return (
    <div style={{ padding: '20px', borderRadius: '8px', background: '#ffffff', boxShadow: '0 4px 6px
rgba(0,0,0,0.05)' }}>
      <h1 style={{ color: '#2c5282', fontSize: '24px', margin: '0 0 10px 0' }}>Hands-on Lab 17: Asynchronous
REST Requests</h1>
      <p style={{ color: '#718096', fontSize: '14px', marginBottom: '20px' }}>
        Objective: Fetch mock data asynchronously from a remote endpoint (api.randomuser.me) inside
componentDidMount.
      </p>
      <hr style={{ border: '0', borderTop: '1px solid #e2e8f0', margin: '15px 0' }} />
      <Getuser />
    </div>
  );
}
```

4. Production Build Verification Logs

To ensure code quality and build compatibility, we executed the production compilation command:
$ npm run build

The build output below verifies that Vite successfully transpiled all JSX modules, resolved stylesheets, and bundled assets into a minimized production distribution package under the /dist folder with zero warnings or errors:

// File: npm run build terminal output

```
> react-deepskilling-apps@0.0.0 build
> vite build

vite v8.1.5 building client environment for production...
 [2K
transforming...[MET] 44 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                   0.47 kB ? gzip:  0.30 kB
dist/assets/index-B1lA3n8u.css    1.80 kB ? gzip:  0.77 kB
dist/assets/index-C3LdQuQr.js   287.78 kB ? gzip: 86.11 kB

[MET] built in 82ms
```